

Lab: Memory Map -- How Your Brain Builds Models of the World

Objective

Explore how your brain constructs and maintains **generative models** (mental maps) of the world, and discover how cognitive biases reveal the limits and shortcuts of those models.

Prerequisites

- Completed modules on Systems, Agents, and Perception
- Read the module on Mental Models

Part 1: Draw Your Mental Map

Goal: Reveal the generative model your brain has built of a familiar place.

Without looking at any references, draw a map of your school from memory. Include hallways, classrooms, the cafeteria, exits, and anything else you remember.

After drawing, compare your map to the actual layout (use a school map or walk the route).

Answer:

1. What parts of your map were accurate?
2. What parts were distorted, missing, or exaggerated?
3. Why do you think your brain emphasized some areas and ignored others?

Write your response here

Part 2: The Bias Spotter

Goal: Identify cognitive biases as "rigid priors" in your mental models.

Read each scenario and identify which cognitive bias is at work. Then explain how it connects to Active Inference.

Scenario	Bias Name	How It Relates to Generative Models
You study only information that confirms what you already believe about a topic.		
You think a coin that landed heads five times in a row is "due" for tails.		
You judge a restaurant as bad because of one bad meal, ignoring ten good ones.		
After hearing the answer to a question, you think "I knew that all along."		

Write your response here

Part 3: Latent States -- What Is Hidden?

Goal: Practice identifying latent (hidden) states that your brain infers but cannot directly observe.

For each situation, identify what is hidden (latent) and what is directly observable (sensory):

Situation	Observable Sensory Data	Latent States (What Your Brain Infers)
A friend is frowning		
You smell smoke in a building		
Your phone buzzes in your pocket		
A teacher pauses mid-sentence		

Write your response here

Part 4: Updating Your Model

Goal: Practice Bayesian updating in everyday reasoning.

Think of a belief you held that changed when you got new evidence. Walk through the update:

1. **Prior belief:** What did you believe before?
2. **New evidence:** What information did you receive?
3. **Updated belief (posterior):** How did your belief change?
4. **Precision:** How confident were you in the new evidence? Did that affect how much you updated?

Write your response here

Discussion Questions

- 1. Why is the phrase "the map is not the territory" important for understanding cognition?
- 2. Can a cognitive bias ever be helpful? Give an example.
- 3. How is a generative model different from a simple memory?

Write your response here

Summary Table

Concept	Definition	Your Example
Generative Model	The brain's internal theory of how the world produces sensory data	
Latent States	Hidden variables the brain infers but cannot directly observe	
Cognitive Bias	A systematic shortcut in the generative model that can lead to errors	
Bayesian Updating	Revising beliefs in light of new evidence	